Bellabeat Smart Device Usage Analysis

Analyzations of Fitbit Activity to advise Bellabeat Marketing Strategy

China Trimble
Data Analytics Case Study
SQL | R | Google Slides
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Business Problem

- **Company:** Bellabeat is a wellness tech firm dedicated to helping women track and improve their health.
- **Objective:** Examine Fitbit data to uncover patterns in how users interact with their smart devices.
- **Goal:** Apply these findings to refine Bellabeat's marketing strategy and product positioning.
- **Product Focus:** The **Bellabeat Time**, which tracks activity, sleep, and wellness.

Data Overview

This analysis us's Kaggle's Fitbit data to examine user activity patterns.

The dataset includes:

- Daily Activity Records
- Step Counts
- Distance Traveled
- Activity Intensity Levels
- Calories Burned

The data tracks user activity from **March to May 2016** from various Fitbit users.

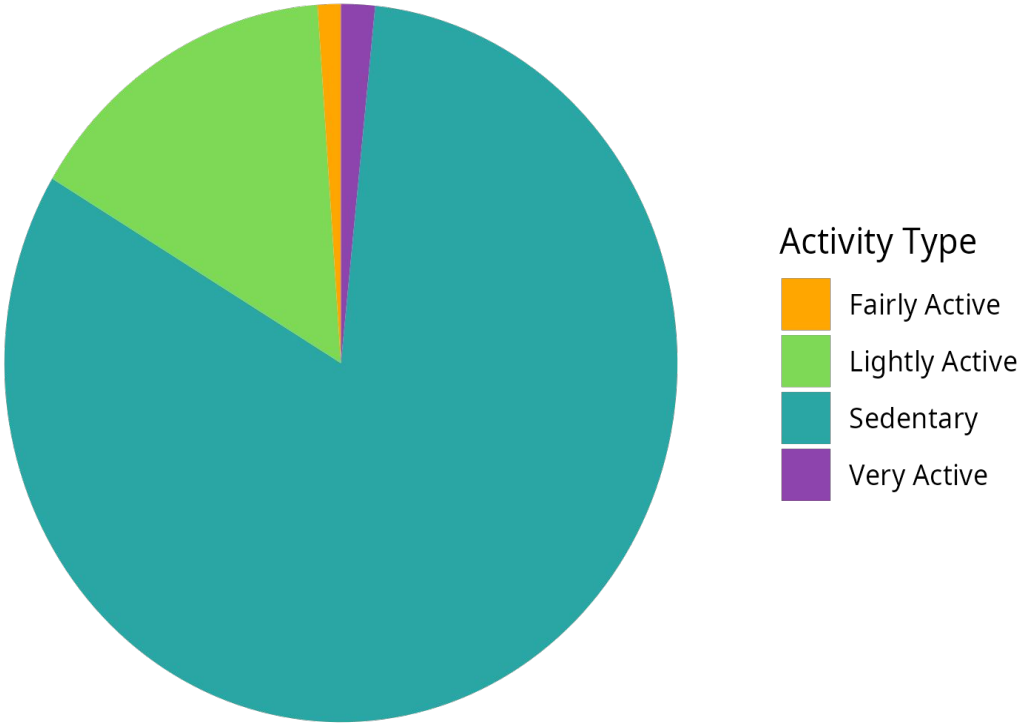
Data Preparation

- Combined two daily activity datasets
- Checked for duplicate records
- Standardized column names
- Verified date ranges
- Removed formatting issues
- Prepared dataset for analysis using **spreadsheets**, **SQL**, and **R**.

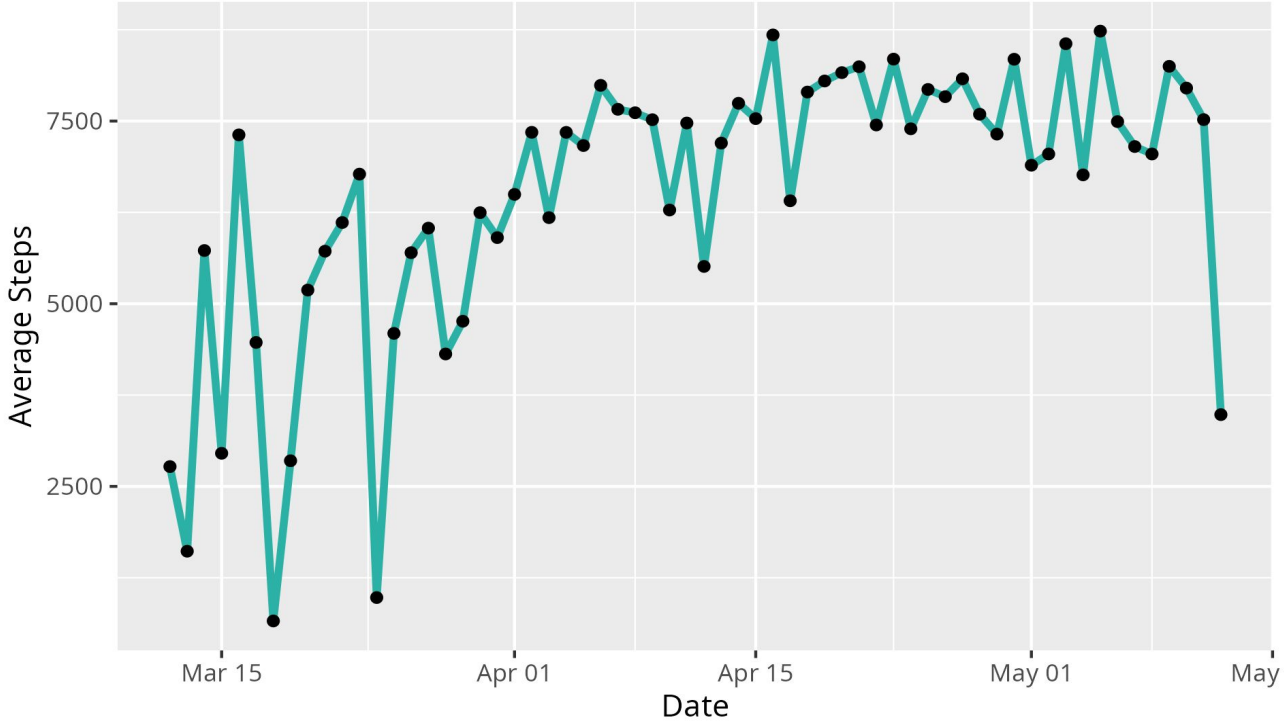
Insight: Users Spend Most Of Their Time Sedentary

A significant part of the day is devoted to sedentary time.

- Most logged minutes are classified as sedentary
- Sedentary behaviors are significantly more prevalent than active bouts.
- Inactivity accounts for a significant percentage of the user's day.



User Activity Trends Overtime



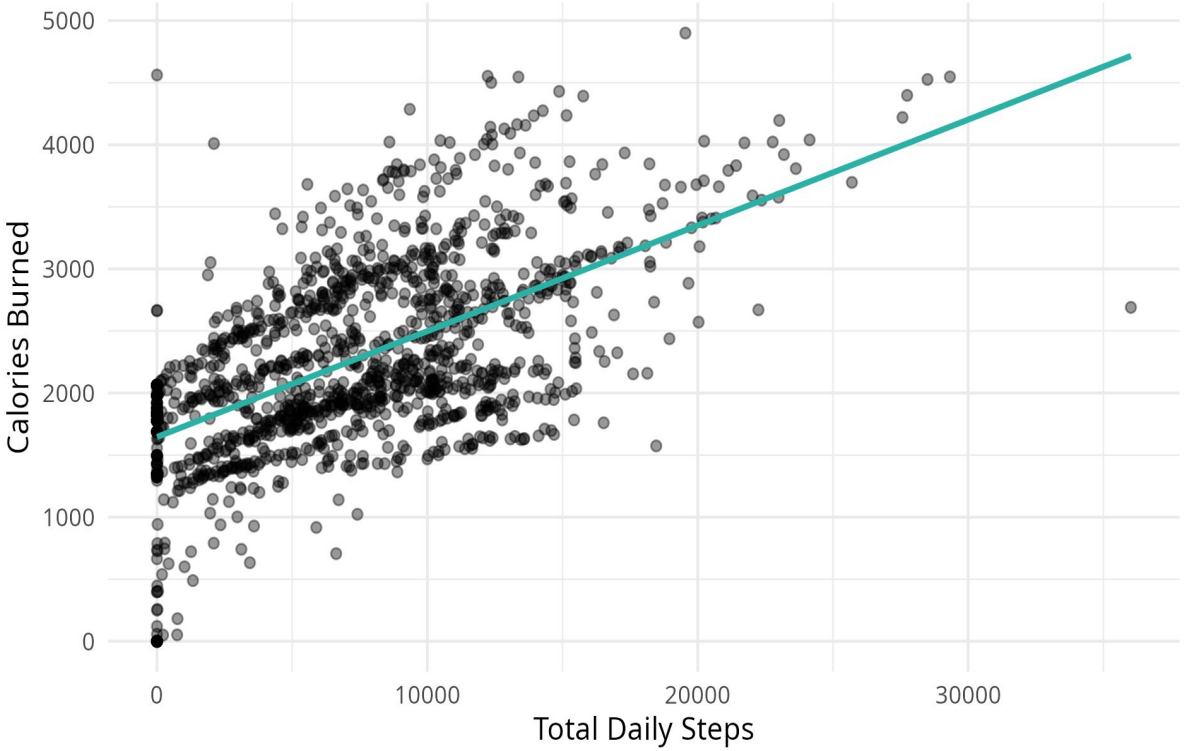
Key Insight:

- Average daily steps **decreased in March.**
- Activity **rose sharply in mid-April.**
- Steps **declined again in late May.**

Possible Explanation:

Seasonal changes or behavioral factors may affect activity levels.

Step Count Positivity Correlates With Calories Burned



Key Insights:

- Daily Step counts and calories burned share a **strong positive relationship.**
- Data shows that users recording **more steps consistently burn more calories**.
- Promoting higher daily step goals is an effective way to boost overall activity.

Recommendations for Bellabeat

- Drive daily movement with **step goals and timely reminders**
- Reduce sedentary time with **inactivity notifications**
- Promote consistent activity habits through structured **daily wellness routines**
- Highlight key product benefits in **marketing campaigns**
- Position the **Bellabeat Time** as a tool for cultivating healthy habits

Totals Used

- **Spreadsheets:** Data cleaning, formatting, and pivot table analysis
- **SQL:** Querying, filtering, and aggregating activity data
- **R:** Data visualization and correlation analysis